STATE OF THE ART SUMMARY — Weight-Bearing Feedback Device

Michael Phelan, Concordia Lutheran High School | Dr. T. Jenkyn, Western University |

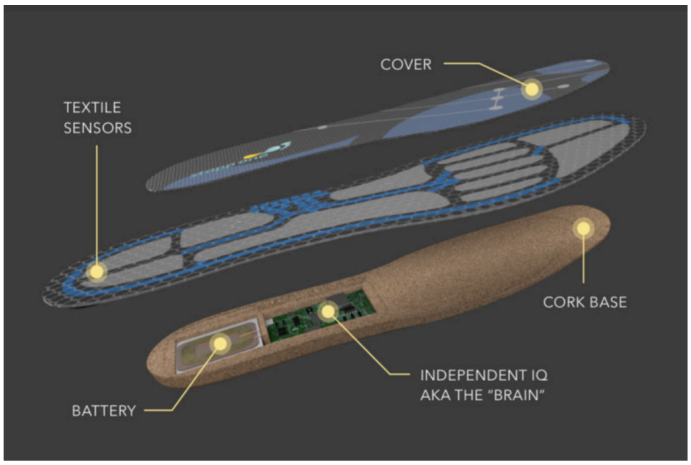
The Problem

- Doctors prescribe partial weight bearing (PWB) following a surgery to help heal bones while promoting early mobilization, transitioning the patient from non-weight bearing to full, unrestricted weight bearing. It is expressed as a percentage of body weight (EX: “only put 20% of your body weight on your right leg”). The percentage would gradually increase as the bone heals. For example: 10% → 20% → 30% → 50% → full
- However, patients at home have no accurate and cost-effective way to measure this. Currently, the main clinical standard is using bathroom scales. The patient would stand on a scale, shift their weight to get a reading, then try to memorize the sensation. However, this leaves a big problem. Standing on a static scale is NOT equivalent to dynamic walking, and sensation cannot be accurately reproduced on walking and day-to-day activities (1).
- Furthermore, compliance is extremely poor. Studies have shown that 27-62% of patients fail to follow their weight-bearing instructions accurately. But this is not a motivation problem; humans literally cannot feel an exact percent of their body weight on a limb (2).

HOWEVER, there are biomedical products out there (besides a bathroom scale) that can give an accurate PWB reading on someone. These products are WAY too expensive and not easily accessible for consumer use. Our goal is to take existing technology and make it cheaper. Let's take a look at a few of existing products, and see how we can apply it to our project:

STAPPONE Rehab

- This is probably the closest existing product to our proposed concept
- It comes as 2 shoe insoles. It has 24 pressure sensor fields and connects to an app.
- Not easily accessible to consumers, need to email for a quote, and currently out of stock. Renting option is available



Sensors:

4×6-axis IMU

4×3-axis acceleration

2×3-axis magnetometer

2×3-axis gyroscope

Internal memory storage: 256 MB

Connection: Bluetooth low Energy 4.2

Battery: 370mAh ~36 hours of use

Sample rates: 50-100Hz

Range: 10-20m

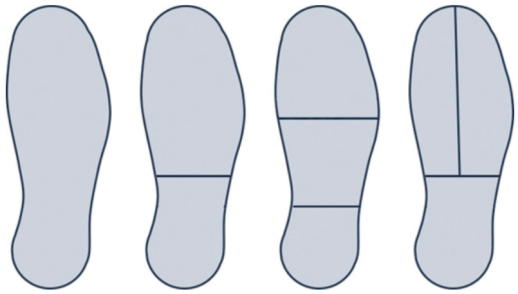
Loadsol by Novel

- Measures load distribution inside the shoe
- Not only for rehab, but can also be used for sports training (golf, soccer, etc.)
- Uses capacitive sensors that cover the entire sole, measures the normal force exerted.
- Notably, it offers different sensor layouts, recording forces in either the whole foot, or targeting different zones. See more below:

Available sensor zone layouts:

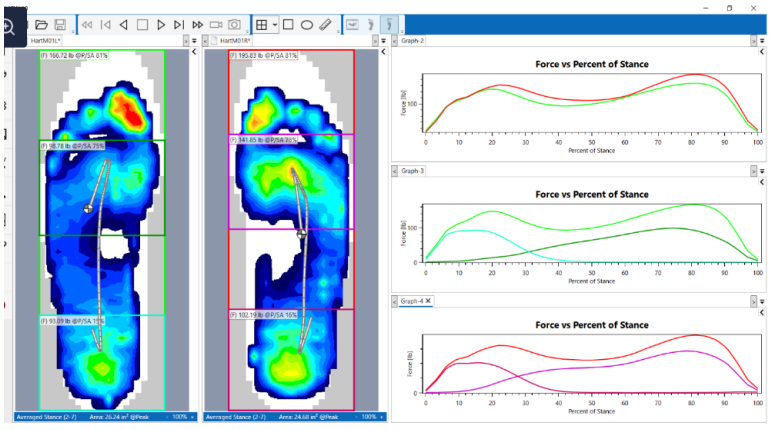
All loadsol are available in 4 different layouts. The more sensor zones, the more detailed you can analyze the force and load distribution. Select

- Version "- t" (total area) for applications where you are only interested in the total static and dynamic force (e.g. partial loading after surgery)
- Version "- ap" (anterior posterior) for applications where the static and dynamic changes of force and loaddistribution between fore-foot and heel are of interest (e.g. running)
- Version "- acp" (anterior, central, posterior) where the force movement from heel to forefoot need to be analyzed in detail (e.g. achillees research)
- Version "- mlp" (medial, lateral, posterior) where the shift of load and force in all directions is of interest (e.g. in golf).



Tekscan F-Scan GO

- I guess you could say this is the "Rolls-Royce" of weight-bearing devices
- Mainly used in research
- Very expensive, over-the-top engineered
- High-resolution pressure distribution system
- Up to 960 individual pressure sensors across the insole
- Provides full spatial pressure map of the foot, not just total force



ComeBack Mobility — Smart Crutch Tips

- Wearable sensors that replace standard rubber crutch tips to monitor and guide weight-bearing during lower-limb rehabilitation
- Algorithm subtracts crutch load from body weight to estimate injured leg loading
- Smartphone app
- Most consumer-accessible option currently available
- However, this only works when patients have crutches.
- Uses force sensitive resistors and load cells



Key findings: the base technology is not that expensive. These companies, such as Tekscan, over complicate things. This research, however, has given key insights on what we can use for our project. Here are some ideas:

Option A: Force Sensitive Resistors

- Thin flexible sensor, resistance decreases as pressure increases
- Directly measure force under the foot
- Can estimate % body weight in real time
- However: non-linear response, needs calibration, and less accurate than clinical sensors



Option B: Conductive/Resistive Foam

- Either conducts electricity (conductive foam) or changes resistance under pressure (resistive foam)
- Same sensing principle as FSR but distributed across foam. It CAN be used as the insole material itself
- See more information I cut from the internet:



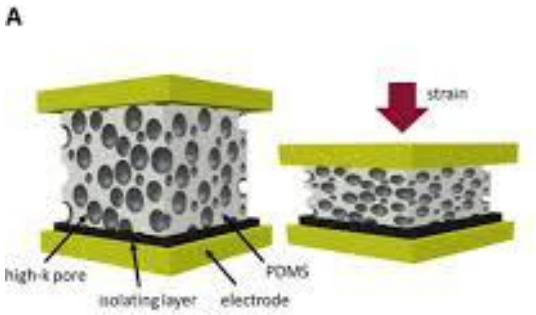
Resting State (High Resistance): In its natural, uncompressed state, conductive foam is full of air gaps, acting as a poor conductor or high-value resistor (often in the Megaohm range).

Compression (Lower Resistance): When force is applied (squished), the air bubbles in the foam are compressed, forcing the conductive material to make more internal contact. This significantly increases the conductive network.

Measurement: This change in resistance can be measured, typically by placing the foam between two conductors (like aluminum foil or copper tape) to create a "sandwich" sensor.

Option C: Capacitive Foam

- This acts as a capacitor. When compressed, the distance between the plates changes, causing the capacitance (ability of an electrical component (a capacitor) to store an electric charge) to change
- Can detect pressure distribution, harder convert to actual force values
- More complex electronics needed to read



to

Option D: Conductive Ink

- A specialized liquid or paste material containing conductive materials
- Resistance changes with deformation
- Used to create custom flexible sensors
- Allows for the creation of flexible, wearable, and printed sensors that can be embedded into smart footwear
- Can measure pressure over larger area



Conclusion: There is no “right way” to create a partial weight bearing device. We can test out different options above, or even use a combination of them (EX: Conductive Ink used with foam).

(1) – https://kar.kent.ac.uk/57924/4/Winter_Are%20current%20methods.pdf

(2) – https://pubmed.ncbi.nlm.nih.gov/24683650/